

Exercise of inheritance - Polymorphism

Exercise: Vehicle System

Create a program to model a vehicle system in a parking lot.

1. Problem description:

- Design an abstract class **Vehicle** with the attributes **brand**, **model**, and **year**, an abstract method **move**, and override the **toString()** method.
- Create two interfaces:
 - **Electric**, with methods: **chargeBattery()** and **batteryLevel()**.
 - **FuelPowered**, with methods: **refillTank()** and **fuelLevel()**.
- Implement the following concrete classes:
 - **ElectricCar**: Inherits from **Vehicle** and implements **Electric**. It will have the attribute **batteryLevel**.
 - **FuelCar**: Inherits from **Vehicle** and implements **FuelPowered**. It will have the attribute **fuelLevel**.
 - **HybridCar**: Inherits from **Vehicle** and implements both **Electric** and **FuelPowered**. It will have the attributes: **batteryLevel** and **fuelLevel**.

2. Restrictions

- When a vehicle is charging its battery, it will display the message: "Charging battery..." and the variable **batteryLevel** will be set to 100.
- When a vehicle is refueling, it will display the message: "Refilling the tank..." and the variable **fuelLevel** will be set to 100.
- Before a vehicle moves, it will check if the **batteryLevel** or **fuelLevel** is greater than 0. Otherwise, it will display the message "cannot move because...". If either **batteryLevel** or **fuelLevel** is positive, the vehicle will display a message like: [Brand] + [Model] is moving using electric power/fuel. After moving, the corresponding level will decrease by 10 units.
- Hybrid cars will consume battery power whenever possible. Only when the battery is depleted will they use fuel.
- The **toString()** method provides a detailed description of each vehicle.

3. Test Class

- In the class **TestVehicles**, declare an array of **Vehicle** that includes instances of **ElectricCar**, **FuelCar**, and **HybridCar**.
- Iterate through the list and invoke the move method repeatedly to verify what happens when the vehicles run out of battery or fuel.
- If a vehicle implements **Electric**, call the methods of that interface. If it implements **FuelPowered**, call the corresponding methods.